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Intelligent aerodynamic modelling method for steady/unsteady flow fields of airfoils driven by flow field images based on modified U-Net neural network

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ABSTRACT

An intelligent modelling method driven by flow field images for predicting steady and unsteady flow field around airfoils has been developed. Signed Distance Field (SDF) images achieve dimensionality enhancement of airfoil geometric information, and 'synthesised images' achieve dimensionality enhancement of the angle of attack of the airfoil and Mach number. An intelligent aerodynamic model for steady flow field of airfoils is constructed based on the U-Net neural network architecture, and further incorporating a long short-term memory (LSTM) module to construct a U-Net-LSTM neural network architecture to extract the temporal features. Typical NACA airfoils results show that, the prediction error for steady flow is less than 1.98%, while the prediction error for unsteady flow is less than 2.56%. Additionally, the model demonstrates good generalization capability, with a generalization error for steady flow less than 2.45% and a generalization error for unsteady flow less than 3.34%. This research provides a new method for intelligent aerodynamic modelling based on physical representations. Compared to existing methods, this method avoids the need for extracting airfoil geometry information and eliminates the necessity of predicting the flow field point by point, making it more concise and efficient.

Highlights

1. An aerodynamic model was constructed using U-Net to rapidly predict the steady flow field around airfoils.
2. A Long Short-Term Memory (LSTM) module was incorporated to capture temporal information, enabling the rapid prediction of the unsteady flow field around airfoils. To address the problem of 'dimension loss' in the modelling datasets, effective data dimensionality enhancement was achieved using SDF images and 'synthesized images'.

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KEYWORDS

Aerodynamic modeling method; steady/unsteady flow fields; airfoils; flow field images; U-Net neural network

1. Introduction

Efficiently obtaining the flow field around airfoils is extremely important (Wang et al., 2024; Wu et al., 2021; Zuo et al., 2024) in the field of aircraft design. Especially in the early stages, engineers needed to iteratively design the airfoil multiple times and quickly evaluate various models to better meet performance requirements. The traditional methods for obtaining the flow field around airfoils can be achieved through three paradigms: experimental science, theoretical science, and computational science. Experimental science involves analyzing the flow field around airfoils through experiments. By observing, measuring, and recording phenomena and data during experiments, salient features of the flow field around airfoils can be obtained. Kamyab Matin et al.

(2017) conducted experimental research on the flow field around the NACA0024 airfoil with steps on the lower and upper surfaces. The experiments utilised particle image velocimetry (PIV) based on instantaneous flow structures to study the flow field around the airfoil at different time intervals. The results disclose that using step in airfoil leads to a decrease in the Strouhal number. In addition, the formation of vortices in wake region and their positions at different times are discussed. However, experimental design usually relies on prior knowledge from experts and requires expensive equipment, significant time, and resources, making it mainly suitable for the later stages of airfoil design. Theoretical science uses mathematical and physical laws to describe and analyze the flow field around airfoils. Researchers establish

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mathematical models to describe the features of the flow field. Tang (2020) focuses on fundamental research of subgrid scale (SGS) models in large eddy simulation (LES). The research explores the impact of grid width variations on the SGS eddy viscosity and introduces a novel SGS model, namely the Subgrid Scale Eddy Viscosity Transport Equation Model (TEM). Applying the TEM model to large eddy simulations of the NACA0012 airfoil. The results show that the TEM model accurately reproduces the SGS kinetic energy distribution with non-equilibrium characteristics and effectively predicts changes in the sound field. Theoretical science establishes the foundation for experimental science and computational science. Computational science combines experimental and theoretical methods by using computer simulations to analyze the flow field. For example, using Computational Fluid Dynamics (CFD) to solve fluid dynamics problems, simulating the performance and flow field features of airfoils using numerical methods. Rao et al. (2021) used CFD technology to calculate the flow fields around symmetrical and asymmetrical airfoils at two different wind speeds, 3 and 15 m/s. They plotted graphs of lift coefficient (CL) and drag coefficient (CD) relative to angle of attack (α), and also observed the effects of geometric shape and angle of attack variation on stall formation. Computational science provides a precise and cost-effective method without the need for large-scale experimental equipment and a large amount of resources. However, in practical engineering projects, as the complexity of the computational objects increases, the number of computational grid points also grows exponentially. Currently, using CFD solvers to compute the flow field requires a significant number of iterative processes, which are both computationally expensive and time-consuming (Shen et al., 2023).

One method for rapidly evaluating the aerodynamic performance of airfoils is to construct a surrogate model (Alizadeh et al., 2020). The basic idea is to perform CFD calculations only on a small number of sampled airfoils. Then, constructing surrogate models using the obtained CFD data. These surrogate models can further be used to compute aerodynamic forces of airfoils beyond the sampled datasets. This method avoids directly solving the Reynolds-averaged Navier-Stokes (RANS) equations, significantly reducing computational time. Currently, surrogate models combined with CFD mainly include multi-fidelity methods, reduced-order models (ROMs), and data-driven models. They have been successfully applied to improve the efficiency of aerodynamic performance assessment, such as polynomial response surfaces (PRS) (Immonen, 2018), artificial neural networks (ANN) (Esfahanian et al., 2024; Li & Chang, 2023), radial basis functions (RBF) (Zhang et al., 2024), support vector

machines (SVM) (Chen et al., 2020), Kriging models (Lee et al., 2018; Xu et al., 2021), and Gaussian process (GP) (Bilal et al., 2020; Luo et al., 2023) regression. However, most traditional surrogate models are only suitable for predicting low-dimensional aerodynamic data, such as lift coefficient, drag coefficient, and moment coefficient, without capturing the high-dimensional flow field features.

With the development of computer science and the growth in datasets scale, big data-based machine learning (ML) techniques (Tamascelli et al., 2024; Xu, 2019) have emerged. The application of deep learning methods (Carta et al., 2021; Gheisari et al., 2023) has facilitated the emergence of a new paradigm for physical system modelling, namely data-driven modelling. Researchers have conducted many studies on using deep learning techniques to rapidly and accurately obtain the flow field around airfoils. Sekar et al. (2019) employed convolutional neural networks to extract geometric information from airfoils. They then inputted the extracted geometric information along with Reynolds number and angle of attack into a feed-forward neural network, thus constructing a model for predicting the flow field. This model can obtain the flow field around airfoils within a matter of seconds. Sun et al. (2021) constructed an efficient deep neural network model to predict compressible flow around transonic airfoils. They utilised the geometric parameters of airfoils and the physical coordinates of flow field points as inputs for the neural network, with the corresponding velocity and pressure at those coordinates serving as outputs, establishing a flow field prediction model for transonic airfoils. Wu et al. (2022) proposed a data-driven reduced-order model based on convolutional neural network and deconvolutional neural network. The model achieved accurate predictions of the flow field around the NACA0012 airfoil and demonstrated good generalisation capability under various aerodynamic conditions. Its prediction process is three orders of magnitude faster than CFD. Chen and Thuerey (2023) proposed a Deep Neural Network (DNN) model based on commonly used structured grids. This model predicts the flow field around airfoils for various aerodynamic conditions, including angle of attack, Reynolds number, and Mach number. For supersonic flow, the model accurately predicts complex flow phenomena at high Reynolds numbers, such as shock/boundary layer interactions, as well as quantitative distributions of pressure coefficients, wall friction coefficients, and total pressure distributions in the wake downstream of the airfoil.

Overall, while traditional aerodynamic modelling has primarily concentrated on steady conditions, recent advancements in intelligent modelling are addressing the

complexities of unsteady aerodynamics. For instance, Chen et al. (2024) have demonstrated the potential of data-driven approaches in estimating aerodynamic loads from sensor data, even in complex gusty environments. Fukami and Taira (2023) have introduced reduced-order modelling techniques that effectively capture the essence of high-dimensional aerodynamic flows, simplifying the representation of extreme vortex gusts. These studies indicate that neural network models can handle unsteady conditions with high accuracy and efficiency. Based on this, this paper first constructs a neural network-based intelligent prediction model for steady flow fields around airfoils, further improves it, and applies it to the prediction of unsteady flow fields.

Regarding the current research on airfoil flow field prediction, the inputs to the aerodynamic models are parameterised geometric information of the airfoil, and the flow field is predicted point by point based on physical coordinates. This method is relatively cumbersome, and the extraction of geometric information of airfoils requires the establishment of a dedicated neural network. Moreover, the cost of predicting the flow field point by point is high and the timeliness is poor. Furthermore, for aerodynamic design, the focus is not only on aerodynamic forces, but also on determining the quality of the design scheme based on the features of the flow field. Therefore, how to reflect the physical laws from the model is the current focus of intelligent modelling. Although physics-informed neural networks (PINNs) embed the physical rules of governing equations, their optimisation process is relatively complex and prone to getting trapped in local optima or leading to unstable training (Mishra & Molinaro, 2023). Additionally, the performance of PINNs heavily depends on the accuracy and completeness of the embedded physical model. If the physical model contains errors or is incomplete, PINNs may fail to correctly capture the true behaviour of the system (Tartakovsky et al., 2020).

To address the above issues, we propose an efficient and accurate model for predicting both steady and unsteady flow field around airfoils. For steady prediction based on the U-Net architecture, the input consists of SDF images of airfoils and 'synthesized images' that combine the airfoil geometry, angle of attack, and Mach number. The output is the velocity contour map of the flow field. This model avoids the need for extracting airfoil geometry information and eliminates the necessity of predicting the flow field point by point, making it more concise and efficient. Simultaneously, due to the unique advantage of skip connections in the U-Net architecture, the model can better extract flow field features, ensuring high prediction accuracy. For unsteady prediction,

incorporating a Long Short-Term Memory (LSTM) module into the U-Net architecture enhances the model's ability to capture temporal variations in the flow field, thereby improving the accuracy of flow field prediction under unsteady conditions. The steady/unsteady aerodynamic model we propose has high prediction accuracy, providing instant feedback during the early stages of airfoil iterative design.

2. Flow field image-driven intelligent aerodynamic modelling method

Conventional aerodynamic modelling methods primarily use aerodynamic data from the aircraft. For example, steady aerodynamic modelling typically involves interpolating and fitting angle of attack-aerodynamic force/moment ($\alpha - C_{l,d,m}$), then predicting aerodynamic data at unknown conditions. Unsteady aerodynamic modelling, on the other hand, involves modelling time-angle of attack-aerodynamic force/moment ($t - \alpha - C_{l,d,m}$) under given motion conditions to predict aerodynamic forces at unknown times or conditions. For nonlinear flow field at large angles of attack, basic data modelling can capture the overall aerodynamic representation of the aircraft but cannot obtain the local variation features of the flow field. The aerodynamic modelling method based on flow field images proposed in this paper addresses this shortage. By using flow field images corresponding to specific states for training, it can predict the flow field under unknown conditions, thereby enhancing the representational significance of the model. In this paper, the aerodynamic modelling method based on flow field images is carried out using neural networks. By constructing a neural network architecture based on image pixel features, it converts geometric information of airfoils into images and performs mixed training with flow field images to predict the flow field under unknown steady/unsteady conditions. The overall modelling process is shown in Figure 1.

2.1. Neural network design

2.1.1. Construction of U-Net architecture based on images of steady flow field

U-Net is a specific convolutional neural network architecture initially proposed by Ronneberger et al. (2015), used for biomedical image segmentation. The standard architecture of U-Net is illustrated in Figure 2. It consists of an encoding path (left side) and a decoding path (right side). The encoding path is a typical architecture of convolutional neural network, consisting of multiple groups of stacked 3×3 convolution. Each convolution is followed by a Rectified Linear Unit (ReLU). ReLU is a

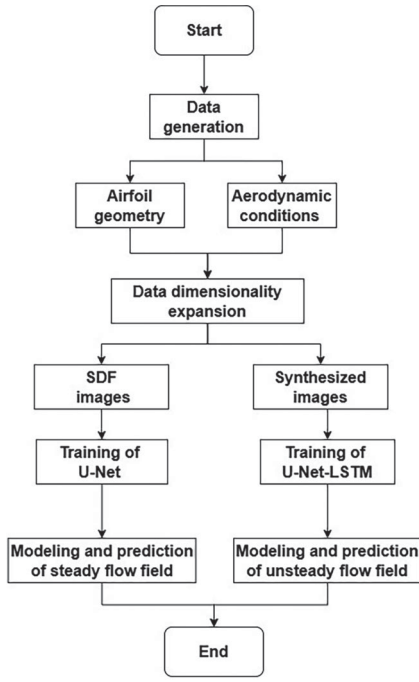


Figure 1. Overall modelling process.

widely used activation function in deep learning, originally proposed by (Nair & Hinton, 2010). The introduction of ReLU marked a significant turning point in the design of activation functions, making it possible to train deeper neural networks. And a 2×2 max-pooling operation with a stride of 2, applied after ReLU. Each module in the decoding path includes upsampling of the feature maps, followed by a 2×2 up-convolution that halves the number of feature channels. This result is concatenated with the feature maps from the encoding path. Then followed by two 3×3 convolutions, with a ReLU activation after each of them. In the final layer, a 1×1 convolution is used to map each 64-channel feature vector to the required number of categories. The standard architecture of U-Net consists of 23 convolutional layers.

In this paper, the standard architecture of U-Net has been modified to make it more suitable for flow field prediction tasks. The modified architecture is shown in Figure 3. In the standard architecture, each convolution changes the height and width of the feature maps. In the modified architecture, zero padding operations are employed to ensure that the height and width of the feature maps remain unchanged after each convolution. This ensures that when concatenating the feature maps obtained from the encoding path with those from the decoding path, no additional cropping operations are required, thereby ensuring the flow field features are not lost. Moreover, in fluid dynamics, boundary conditions

play a decisive role in the evolution of the flow field, and neglecting boundary details may lead to a loss of model accuracy. By using zero padding at the boundaries of input data, convolutional layers can retain more boundary information during the convolution process, thereby improving modelling performance. Compared to other padding methods such as edge padding, reflection padding, and replication padding, zero padding ensures boundary symmetry without introducing new information, allowing neural networks to effectively capture flow characteristics, especially in complex regions near fluid boundaries. Specifically, if the size of input feature map is $H \times W$, the size of convolution kernel is $K \times K$, the size of padding is P , and the size of stride is S , then the size of out feature map is (Gu et al., 2018):

$$H_{out} = \frac{H - K + 2P}{S} + 1 \quad (1)$$

$$W_{out} = \frac{W - K + 2P}{S} + 1 \quad (2)$$

When P takes a specific value, the size of output feature map remains the same as the input, ensuring that the flow field features are not lost.

Batch Normalization (BN) are added after each convolutional layer to accelerate the training process of the neural network, while also enhancing its performance and stability (Ioffe & Szegedy, 2015). For flow field images, the distribution of flow field information varies significantly due to differences in position and scale. BN can normalise the distribution of each feature channel, helping the network converge faster and improving the generalisation ability. The formula for BN is as follows (Garbin et al., 2020):

$$BN(x) = \gamma \frac{x - \mu}{\sigma} + \beta \quad (3)$$

In this formula, x is the input feature, μ and σ are the batch mean and standard deviation respectively, γ and β are the learnable scaling and shifting parameters.

Additionally, the transpose convolutions in the encoding path are replaced with bilinear interpolation upsampling. Efficiency is an important consideration in flow field prediction tasks, especially when dealing with high-resolution images or datasets. The computational cost of bilinear interpolation upsampling is relatively low. It calculates the new pixel value by averaging the weighted values of the four nearest neighbouring pixels. This method is numerically stable and less prone to causing artifacts or oscillations. Transposed convolutions may cause ‘Checkerboard Artifacts’ in some cases, which manifest as checkerboard patterns in the upsampled results. Bilinear interpolation upsampling tends to be smoother and

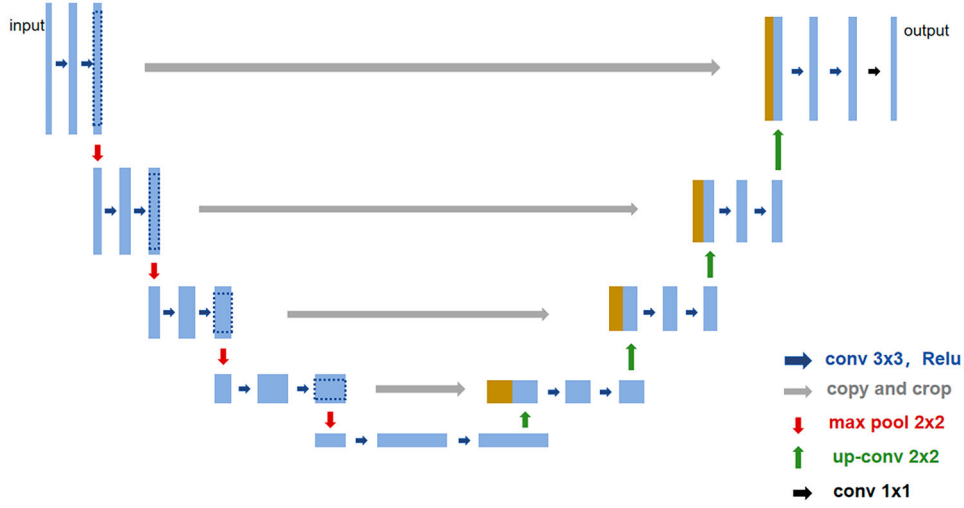


Figure 2. Standard architecture of U-Net.

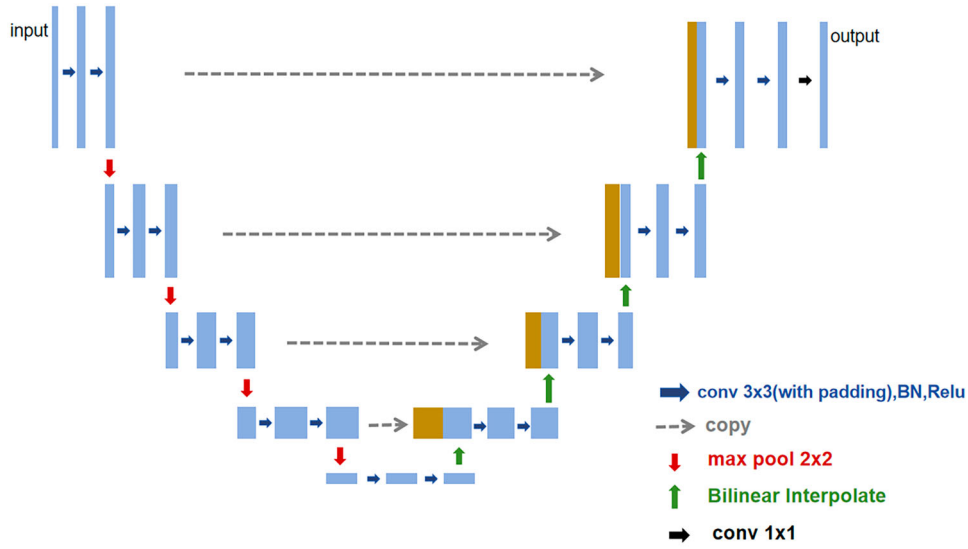


Figure 3. Modified architecture of U-Net.

can help avoid this issue (Wang, 2022). It can better preserve details and spatial consistency to a certain extent, which is crucial for preserving accurate spatial information in flow field prediction tasks. The formula for bilinear interpolation upsampling is as follows (Kumar et al., 2023):

$$\begin{aligned} \text{Interpolation}(x, y) = & (1 - x)(1 - y)Q_{11} + x(1 - y)Q_{21} \\ & + (1 - x)yQ_{12} + xyQ_{22} \end{aligned} \quad (4)$$

In the formula, Q_{11} , Q_{12} , Q_{21} , Q_{22} are the values of the four nearest neighbouring pixels, x and y are the coordinates of the target position.

2.1.2. Construction of U-Net-LSTM architecture based on images of unsteady flow field

To predict the unsteady flow field around airfoils, an LSTM module is embedded into the U-Net architecture. LSTM networks, a special type of recurrent neural network (RNN) primarily used for processing and predicting time series data, were first introduced by (Hochreiter & Schmidhuber, 1997). This integration aims to better capture the changes of the flow field in the temporal dimension. The deep stacked U-Net-LSTM architecture is shown in Figure 4. An LSTM module is added after each convolutional layer to extract multi-scale temporal features. These features are combined with the output of the corresponding upsampling layer and then input

into the next upsampling layer. The width and height of the output image from the final upsampling layer are the same as those of the individual input image.

The specific structure of this neural network are as follows:

1. Input: Assuming there are $i+j+1$ images in the datasets, i.e. $[p_1, p_2, \dots, p_{i+j}, p_{i+j+1}]$, the image size is $w \times h$. The $i+1$ image is considered as the ground truth. According to this assumption, the datasets can be divided into $j+1$ samples, i.e. $[p_1, p_2, \dots, p_i/p_{i+1}]$, $[p_1, p_2, \dots, p_{i+1}/p_{i+2}]$, $\dots$, $[p_j, p_{j+1}, \dots, p_{i+j-1}/p_{i+j}]$, $[p_{j+1}, p_{j+2}, \dots, p_{i+j}/p_{i+j+1}]$, where the first j samples are the training datasets.
2. Encoder: Taking $j=1$ as an example, an LSTM module is added after each convolutional layer and activation function in the encoder. The size of the convolutional kernel is 3×3 . The input of each LSTM is a multi-scale feature of i time steps, and the output can be chosen as either an sequence of i time steps, with a size of $i \times W \times H \times 1$, or only the last time step, with a size of $1 \times W \times H \times 1$.
3. Decoder: Taking LSTM only outputting the last time step as an example, the process of decoding consists of four upsampling layers, convolutional layers, and activation functions. Before the first convolution operation of the encoder, the input image is connected to the last upsampling layer through LSTM. After upsampling, the output of the last convolutional layer of the encoder is used as the input for subsequent upsampling layers through LSTM. Depending on the dimension and size of the input features, the number of neurons in the hidden layer and the form of activation functions can be adjusted.

The length of the time series determines the time step parameter of the LSTM.

4. Output: Before prediction, multiple convolutional layers and activation function layers can be appropriately added to further extract temporal information. The neural network model learns the evolution information of the first N time steps and predicts the changes of flow field images from the N th time step to the $N+1$ th time step, with a size of $W \times H \times 2$.

Pseudocode implementation:

Class DownBlock:

Initialise(conv_kernels, lstm_kernels, stride, data_format):

For each (kernel, filters) in conv_kernels:

Initialise Conv2D layer with kernel parameters

Initialise BatchNormalization layer

Initialise LeakyReLU activation

For each (kernel, filters) in lstm_kernels:

Initialise LSTM layer with filters and kernel parameters

Call(inputs):

Apply Conv2D layers sequentially on inputs

Apply BatchNormalization and LeakyReLU activations

Prepare input for LSTM by reshaping (e.g. [batch_size, time_steps, height * width, channels])

For each LSTM layer:

Apply LSTM layer on reshaped inputs

Reshape the output back to (batch_size, height, width, channels)

Return processed tensor and skip connection

Reset_states_per_batch(is_last_batch):

Update states of LSTM layers based on is_last_batch

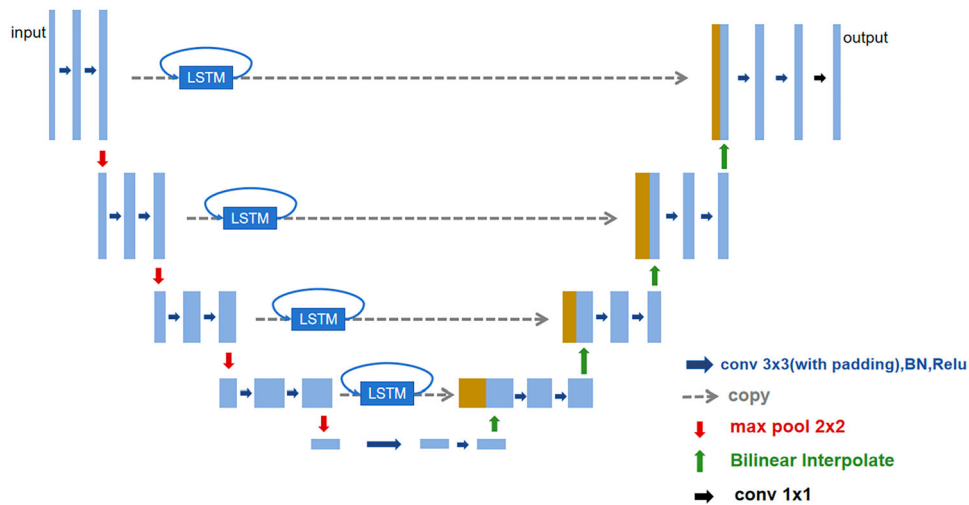


Figure 4. Architecture of U-Net-LSTM.

```

Get_states():
    Return states of all LSTM layers
Set_states(states):
    Set states of LSTM layers from provided states
Class UpBlock:
    Initialise(kernels, up_factor, data_format, return
_logits):
        For each kernel in kernels:
            Initialise Conv2D layer with kernel param-
eters
            Initialise LeakyReLU activation
Call(inputs):
    Upsample input_sequence
    Concatenate upsampled input_sequence with
skip connection
    For each Conv2D layer and LeakyReLU activa-
tion:
        Apply Conv2D layer
        Apply LeakyReLU activation
    Return processed tensor
Class ULSTMnet2D:
    Initialise(net_params, data_format, pad_image):
        Create DownBlock instances for each down
layer
        Create UpBlock instances for each up layer
        Initialise Conv2D layer for flow prediction
Call(inputs):
    For each DownBlock layer:
        Pass inputs through DownBlock layer
    For each UpBlock layer:
        Pass data through UpBlock layer with skip
connections
        Compute flow and apply SpatialTransformer
    Return transformed output and flow
Reset_states_per_batch(is_last_batch):
    Reset states for all DownBlock layers
Get_states():
    Return states from all DownBlock layers
Set_states(states):
    Set states for all DownBlock layers

```

2.1.3. Loss function

The loss function for the neural network model in this paper is defined as follows:

$$\varsigma_{mse} = \frac{1}{n(p)} \sum_{p=1}^{n(p)} \left(\frac{1}{3} \sum_{j=1}^3 (y_{pj} - \hat{y}_{pj})^2 \right) \quad (5)$$

In this function, p is the set of all pixels in the image, $n(p)$ is the number of pixels in the image, j is the cyclic index of the three channels of the RGB image, $\hat{y}_{pj}$ is the predicted value at the channel j for the pixel p , and y_{pj} is the corresponding label.

2.2. Design of modelling datasets based on dimension matching

Unlike traditional aerodynamic models, the aerodynamic model based on flow field images requires data that includes time, angle of attack, Mach number, and flow field images. It is evident that flow field images are high-dimensional data (with a large data volume), while time, angle of attack, and Mach number are low-dimensional data (with a small data volume). The dimensional difference between them can lead to the phenomenon of 'dimensional loss' during model training, where features extracted from high-dimensional data overwhelm the low-dimensional data, making it difficult to extract information from the low-dimensional data.

With recent advances in neural networks, it has been shown that combining different types of neural networks can effectively handle data of varying dimensions. For example, Morimoto et al. (2021) successfully achieved dimensionality reduction and effective modelling in fluid flow analysis by combining MLP and CNN.

In this study, we chose to adopt a dimensionality enhancement strategy to address the issue of information loss in low-dimensional data during the training process. By increasing the dimensions of low-dimensional data such as time, angle of attack, and Mach number to match the dimensions of flow field images, we ensured that the model could more effectively extract information from these variables. Specifically, we synthesised the airfoil geometry, angle of attack, and Mach number into a 'synthetic image', which, along with the airfoil's SDF images, served as the input to the model, with the airfoil's velocity cloud map as the output.

Compared to handling different dimensional data through neural networks, which typically requires large amounts of data to train the network parameters and obtain effective mappings, the dimensionality enhancement method used in this paper is a data preprocessing strategy. It ensures dimensional consistency at the data input stage, avoiding the need for complex transformations within the neural network architecture. The training process of the aerodynamic model is illustrated in Figure 5.

2.2.1. Dimensionality expansion of airfoil geometry based on SDF images

The velocity field around airfoils is related to the airfoil geometry. Therefore, it is crucial to transform the airfoil geometry into a form that is effective for neural network training. Many studies have demonstrated that the effectiveness of SDF as a representation of geometric information for neural network training. For instance, Ribeiro et al. (2020) used SDF to represent the distance

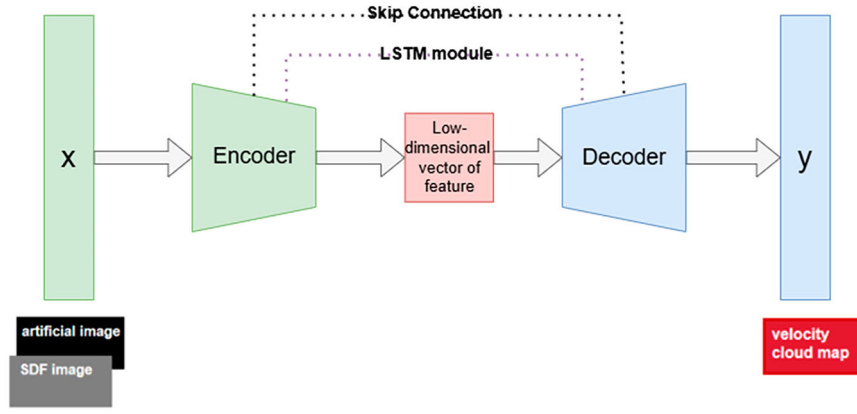


Figure 5. Training process of the aerodynamic model.

from the surface of a geometric object to individual points in a CFD grid as input for a convolutional neural network model, enabling efficient prediction of non-uniform steady-state laminar flows. Chen and Akasaka (2021) proposed a 3D convolutional neural network model for rapid prediction of the flow field around vehicles, where SDF images of vehicle geometries were used as inputs, with the output being the 3D flow field. To calculate the SDF, we first create a zero level set, which is a set of points used to determine the boundary (surface) of the geometry in the domain, as shown in formula 6:

$$Z = \{(i, j) \in R^2 : f(i, j) = 0\} \quad (6)$$

In this formula, f is the level set function, which is $f(i, j) = 0$ when and only when (i, j) is on the geometric boundary, $f(i, j) < 0$ when and only when (i, j) is inside the geometry, and $f(i, j) > 0$ when and only when (i, j) is outside the geometry.

The definition of the $D(i, j)$ related to the $f(i, j)$ is as follows (Lin et al., 2020):

$$D(i, j) = \min_{(i', j') \in Z} |(i - j) - (i' - j')| \text{sign}(f(i, j)) \quad (7)$$

In this formula, $D(i, j)$ is a directed distance function that measures the distance from a given point (i, j) to the nearest boundary of the closed geometric shape Z , and its sign is determined by whether the point is inside or outside the geometry.

Compared to geometric parameters, the SDF provides a universal representation for different geometric shapes and can be effectively combined with neural networks. Therefore, in this paper, the SDF sampled on a Cartesian grid is used as the geometric representation of airfoils.

This paper focuses on the NACA Four-Digit Series Airfoils, developed by the National Advisory Committee for Aeronautics (NACA). Each airfoil is designated

by the letters 'NACA' followed by four numbers, which represent its geometric parameters. By substituting these numbers into specific equations, the geometry of the airfoil can be determined.

The meaning of the four digits:

NACA XYZZ

X – Relative camber

Y – Maximum camber position

ZZ – Relative thickness

For example, NACA 2412 indicates that the airfoil has a relative camber of 2%, the maximum camber position is at 40% of the chord length, and the relative thickness is 12%.

The NACA0012 airfoil and its SDF image are shown in Figure 6.

2.2.2. Dimensionality expansion of angle of attack and mach number based on 'synthetic images'

Zhang et al. (2018) proposed the concept of 'synthetic images', which combine the airfoil geometry, angle of attack, and Mach number into a single data format named 'synthetic images'. These 'synthetic images' are input into a convolutional neural network to predict the aerodynamic coefficients of airfoils. The results show that these 'synthetic images' can serve as an effective data format for neural network training, integrating the geometry of airfoil, angle of attack, and Mach number efficiently. We adopt this approach, using the NACA0012 airfoil as an example, with an angle of attack range of $[-10^\circ, 10^\circ]$ and a Mach number range of $[0.1, 0.6]$. A geometry image of the airfoil is generated for each 1° interval of the angle of attack. According to the transformation formula, the airfoil geometry and angle of attack are combined with the Mach number to generate six 'synthetic images' (grayscale) for each Mach number from 0.1 to

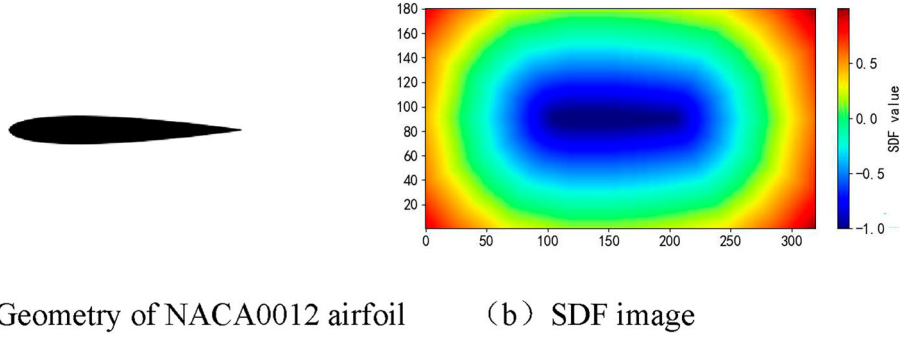


Figure 6. Geometry of airfoil and its SDF image.

Table 1. Selection rules for airfoils and computational conditions for cases.

		Min	Max	Step
Airfoil parameters	Max camber (%)	0	5	1
	Maximum camber position (0.1)	0	5	2
	Max thickness (%)	5	15	5
Flow conditions	Mach number	0.1	0.6	0.1
	Angle of attack	-10	10	1

0.6. The transformation formula is as follows:

$$\bar{Z} = \left(1 - \frac{Z}{Z_{\max}}\right) \frac{M_{\infty}}{M_{\infty, \max}} \quad (8)$$

In this formula, Z is the value of each pixel in the airfoil images, ranging from $[0, 255]$; $Z_{\max}$ is the maximum value of each pixel in the airfoil images; $\bar{Z}$ is the value of each pixel in the 'synthetic images', ranging from $[0, 255]$; M_{∞} is the Mach number of freestream; $M_{\infty, \max}$ is the maximum Mach number in the datasets, which is 0.6 in this paper.

3. Performance validation of steady/unsteady aerodynamic model

3.1. Datasets preparation

For the steady conditions, We selected 33 NACA airfoils with an angle of attack range of $[-10^\circ, 10^\circ]$ and a Mach number range of $[0.1, 0.6]$. For each airfoil, five angles of attack and two Mach numbers were randomly selected within the ranges. This resulted in the generation of ten flow cases for each airfoil. Therefore, a total of 330 cases were created for the 33 airfoils. The selection rules for the airfoils and the computational conditions for the cases are shown in Table 1.

For the unsteady conditions, the same 33 NACA airfoils were selected according to the above rules. The research focused on pitch oscillation motion around the quarter-chord point of the airfoil. The variation of angle of attack is as follows:

$$\alpha = 2^\circ + 6^\circ * \sin(\omega t) \quad (9)$$

In the formula, ω is the oscillation frequency, $k = \frac{\omega C}{2V}$ is the reduced frequency, C is the reference length (chord length of the airfoil), and V is the velocity.

The N-S equations, constructed based on the three fundamental laws of conservation of mass, momentum, and energy, are currently the most comprehensive equations for describing the laws of fluid motion. However, due to the highly nonlinear nature of the N-S equations, Direct Numerical Simulation (DNS) is extremely challenging. Existing CFD methods simplify the N-S equations with various assumptions and then close the equations using turbulence models. Based on the Morkovin hypothesis (1962), the Reynolds-Averaged N-S equations (RANS) treat turbulence as a superposition of time-averaged flow and fluctuating flow, while neglecting density fluctuations. The RANS method applies time-averaging to the governing equations and represents the effects of fluctuating flow using turbulence models. The Unsteady Reynolds-Averaged Navier-Stokes equations (URANS) (Lanade et al., 2023) for unsteady fluid motion are:

$$\frac{\partial \bar{\rho}}{\partial t} + \frac{\partial}{\partial x_i} (\bar{\rho} \bar{u}_i) = 0 \quad (10)$$

$$\frac{\partial}{\partial t} (\bar{\rho} \bar{u}_i) + \frac{\partial}{\partial x_j} (\bar{\rho} \bar{u}_i \bar{u}_j) = \frac{\partial \bar{\rho}}{\partial x_i} + \frac{\partial}{\partial x_j} (\bar{\tau}_{ij}) - \bar{\rho} \bar{u}_i'' \bar{u}_j'' + S_m \quad (11)$$

$$\frac{\partial}{\partial t} (\bar{\rho} E) + \frac{\partial}{\partial x_i} (\bar{\rho} \bar{u}_i H) = \frac{\partial}{\partial x_i} \quad (12)$$

$$\left[-\mu C_p / \text{Pr} \frac{\partial T}{\partial x_i} - \bar{\rho} \bar{u}_j'' h'' + \bar{\tau}_{ij} u_j'' - \frac{1}{2} \bar{\rho} \bar{u}_j'' u_i'' u_i'' \right] + \frac{\partial}{\partial x_j} [\bar{u}_i (\bar{\tau}_{ij} - \bar{\rho} \bar{u}_j'' \bar{u}_i'')] + \bar{\rho} Q \quad (13)$$

In the formula: ρ represents the fluid density; $\bar{\rho}$ denotes the average of ρ ; u_i , u_i'' , and $\bar{u}_i$ represent the absolute velocity in the i direction, the fluctuating velocity,

and the time-averaged velocity, respectively; τ_{ij} represents the shear stress; h represents the enthalpy; E represents the total energy; H represents the total enthalpy; P represents the atmospheric pressure; T represents the ambient temperature; μ represents the dynamic viscosity coefficient; Pr represents the Prandtl number; S_M represents the momentum source term; S_E represents the energy source term.

To solve the Unsteady Reynolds-Averaged Navier-Stokes (URANS) equations, turbulence models must be introduced, as the URANS equations involve turbulence stress terms after time averaging, which need to be described by turbulence models to achieve closure. Common models, such as the $k-\varepsilon$ model, $k-\omega$ model, and Large Eddy Simulation (LES), can provide appropriate solutions based on the requirements of different flow scenarios. The $k-\varepsilon$ model is used in this paper.

The two-dimensional benchmark flow simulation of an airfoil is conducted using air as the working fluid. The boundary conditions include a velocity inlet at the entry, a pressure outlet at the exit, and fixed no-slip conditions on the wall boundaries. The momentum, turbulent kinetic energy, and turbulent dissipation rate are all discretized using a second-order upwind scheme. Appropriate convergence criteria are set to simulate the flow around the airfoil at different angles of attack.

3.2. Performance validation of steady aerodynamic model

We employ the U-Net architecture shown in Figure 2 for modelling and predicting steady flow field. The SDF images of airfoils and ‘synthetic images’ are used as input data for the aerodynamic model, while velocity cloud maps are the output data. 330 groups of cases are divided

into training, validation, and test datasets in a ratio of 8:1:1. During training, the model is considered to have converged when the loss function changes very little or stops decreasing after several iterations, and the validation error stabilises. Specifically, if the change in the loss value is less than $1e-5$ and there is no significant fluctuation in the validation loss for 10 consecutive epochs, the model is deemed to have converged, and the training is terminated. This criterion ensures that the model does not continue unnecessary training after reaching optimal performance, thus avoiding overfitting and improving computational efficiency. To visualise the training process, the error convergence curve of the training is shown in Figure 7. It can be observed that the convergence of this process occurs around the 300th epoch, while the decrease in loss around the 200th epoch is due to the decay of the learning rate. Although the predicted flow field images shown at the 250th epoch may appear blurry, the overall flow field structure has been captured. By the 350th epoch, the predicted flow field images closely resemble the ground truth, and further iterations continue to refine the predicted flow field images.

Figure 8 illustrates the prediction results of the steady aerodynamic model on the test datasets, compared with the ground truth. The left column is the ground truth images, the middle column is the predicted images, and the right column is the error. The error calculation formula is as follows:

$$Error(P_n, \tilde{P}_n) = \frac{1}{N} \sum_{(u,v) \in \omega} ||P_n(u, v) - \tilde{P}_n(u, v)||^2 \quad (14)$$

In this formula, p_n is the n th ground truth image, $\tilde{p}_n$ is the n th predicted image, N is the total number of pixels in the image, ω is the set of pixel coordinates, and (u, v) is a pixel coordinate.

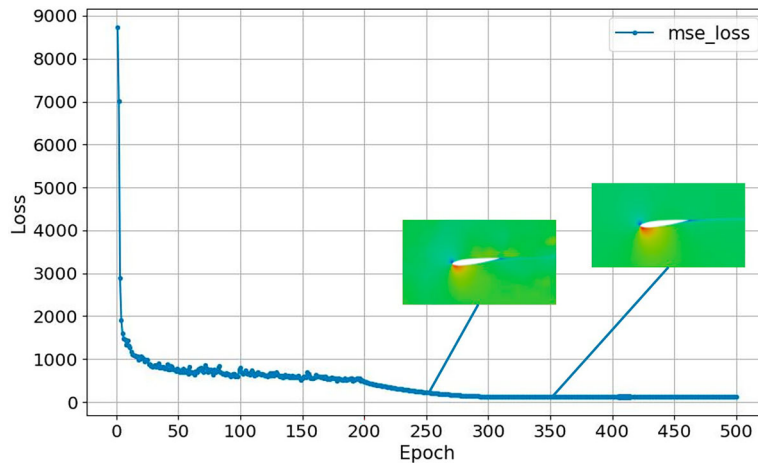


Figure 7. Error convergence curve.

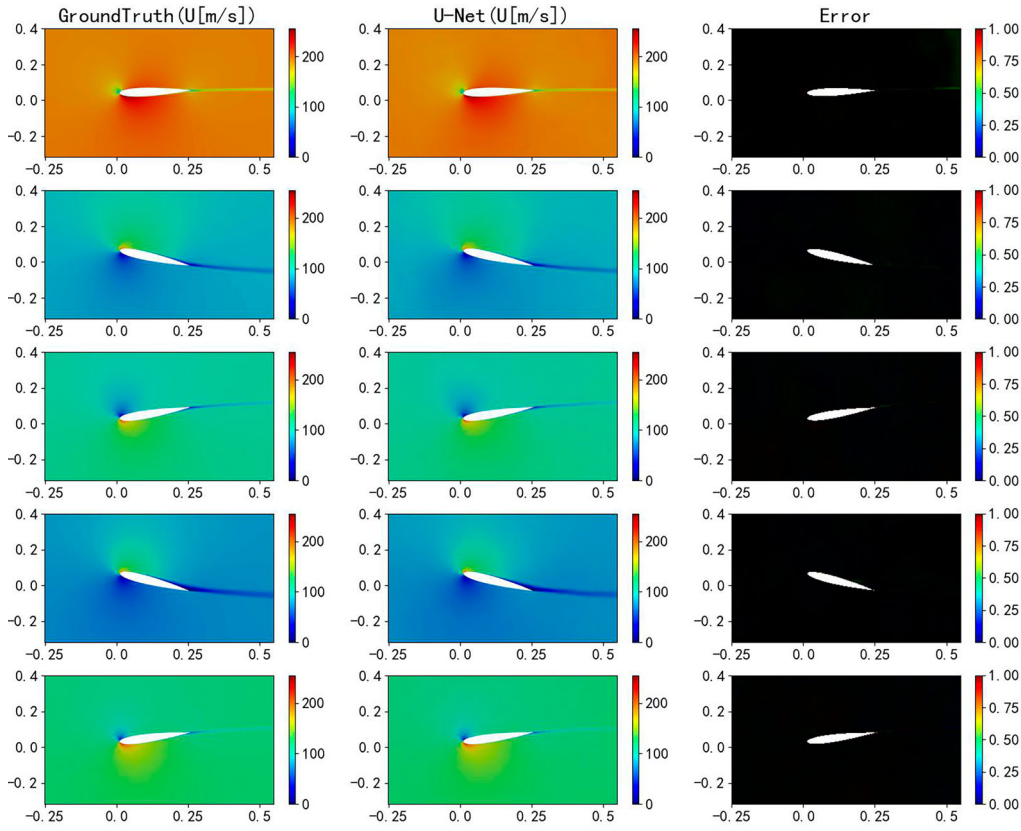


Figure 8. Ground truth and predicted results.

The maximum prediction error is 1.98%. It is evident that the aerodynamic model effectively captures the features of steady flow fields and predicts the steady flow fields at low Reynolds numbers with high accuracy.

3.3. Performance validation of unsteady aerodynamic model

We employ the U-Net-LSTM architecture shown in Figure 4 for modelling and predicting unsteady flow field. 528 groups of cases are divided into training, validation, and test datasets in an 8:1:1 ratio. Each case consists of a series of time sequence images of airfoil pitching oscillation, with a time interval of 0.01s. The neural network takes the first N time sequence images from each case as input and predicts the subsequent time sequence images.

During training, if the change in the loss value is less than $1e-5$ and there is no significant fluctuation in the validation loss for 10 consecutive epochs, the model is considered to have converged, and training is terminated. As shown in Figure 9, it illustrates the time sequence images of NACA0012 airfoil under the condition of $k = 0.08$ at $t = 0.01$ s, 0.05 s, and 0.09 s during pitching oscillation.

Figure 10 illustrates the prediction results of the unsteady aerodynamic model on the test datasets, compared with the ground truth. Based on the prediction results at different time steps, it can be concluded that the U-Net-LSTM architecture constructed in this paper exhibits good adaptability to unsteady time sequence data, with a maximum prediction error of 2.56%. It is evident that the unsteady aerodynamic model effectively

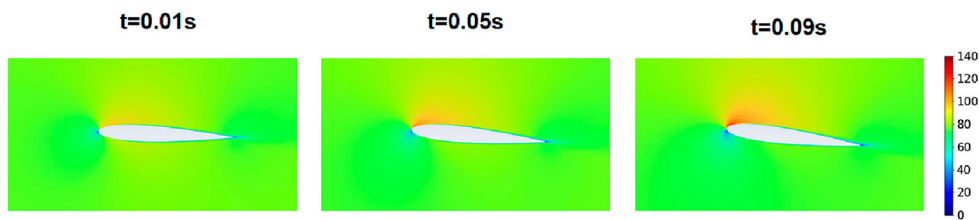


Figure 9. Time sequence images of pitching oscillation of NACA0012 airfoil.

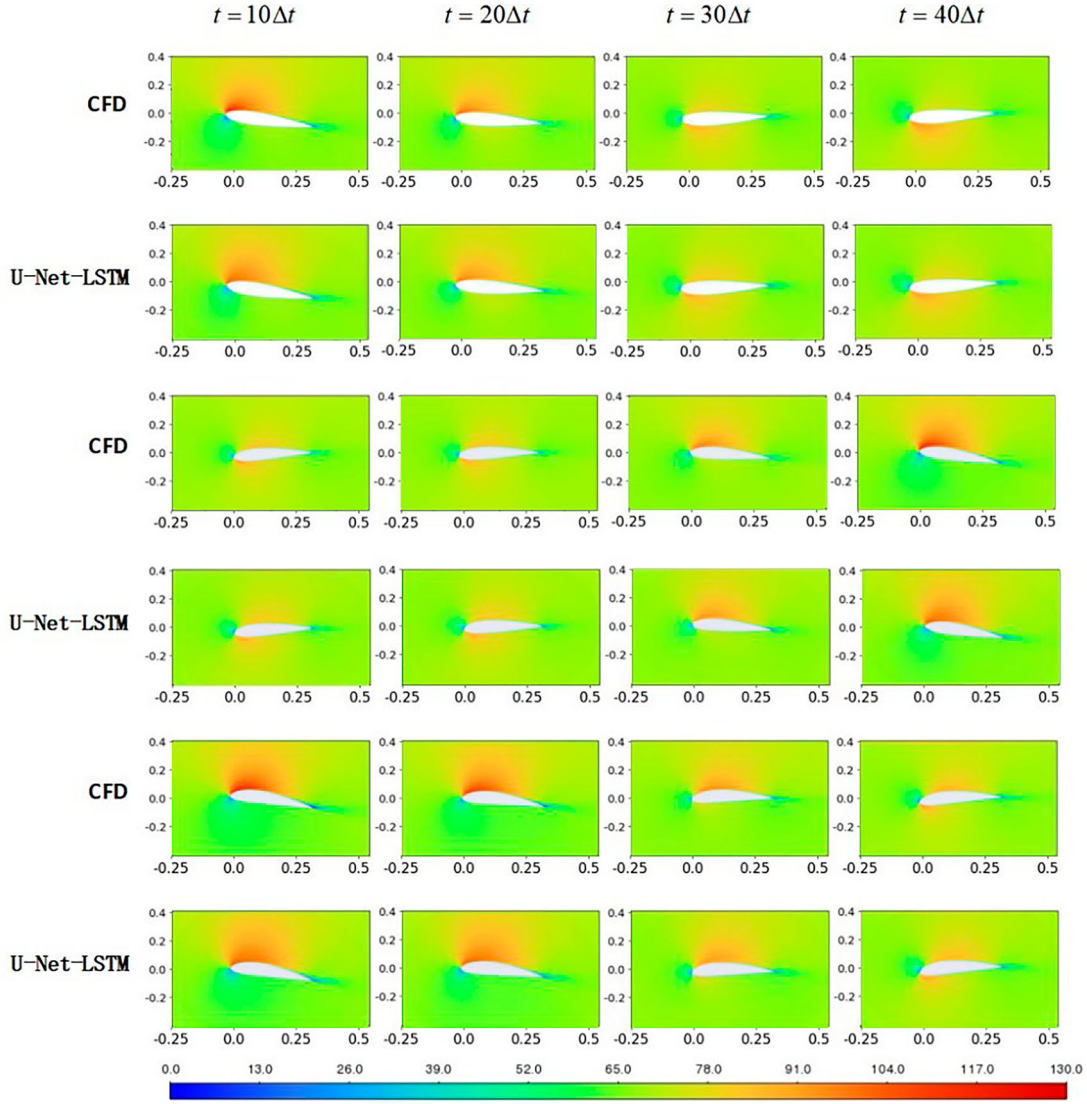


Figure 10. Ground truth and predicted results.

captures the temporal variation features of unsteady flow fields and predicts the pitching oscillation flow field with high precision.

4. Generalisation performance of steady/unsteady aerodynamic model

4.1. Generalisation performance of steady aerodynamic model

Testing the generalisation performance of the steady aerodynamic model. Select five airfoils that did not appear in the training datasets, i.e. NACA0007 NACA 0012, NACA1408, NACA2412, and NACA4412, under the conditions of attack angle of 6.5° and Mach number of 0.45, input the SDF images and ‘synthetic images’

into the trained steady aerodynamic model. It should be emphasised that the steady aerodynamic model has not been trained for these specific conditions. The predicted results are shown in Figure 11, with a maximum prediction error of 2.45%. It is evident that the steady aerodynamic model has high generalisation capability, accurately predicting flow fields beyond the training datasets.

4.2. Generalisation performance of unsteady aerodynamic model

Testing the generalisation performance of the unsteady aerodynamic model. Select three airfoils that did not appear in the training datasets, i.e. NACA0012 NACA 2412, and NACA4412, under the conditions of $k = 0.075$, input the time sequence SDF images and ‘synthetic

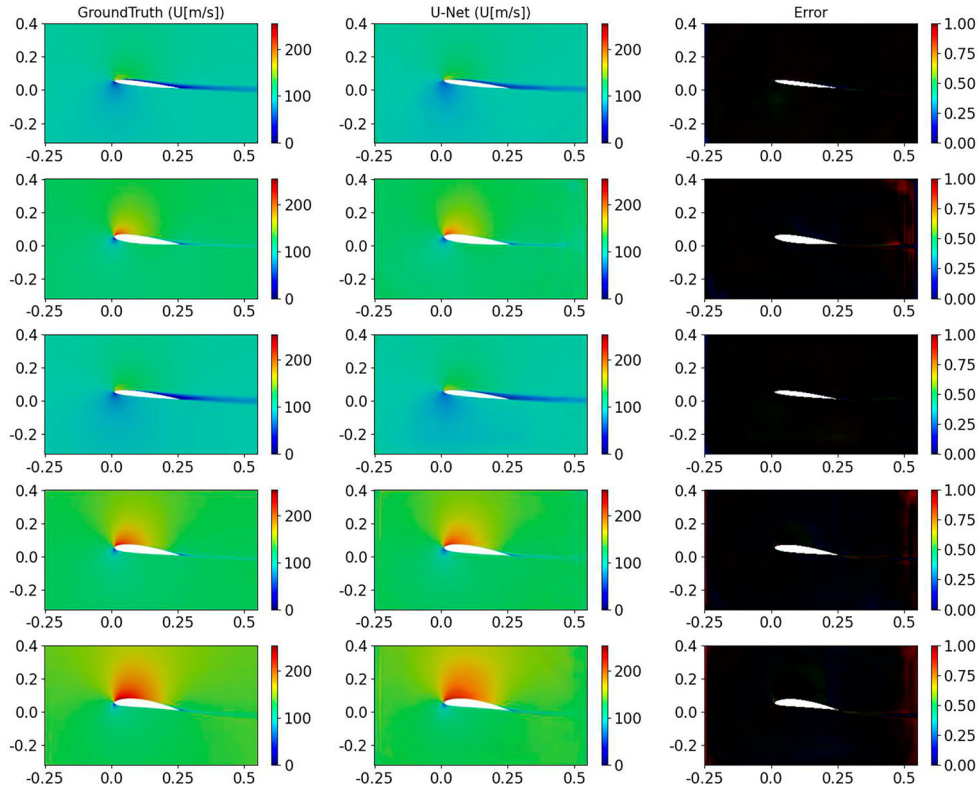


Figure 11. Ground truth and predicted results.

images' into the trained unsteady aerodynamic model. Same as above, the unsteady aerodynamic model has not been trained for these specific conditions. The predicted results are shown in Figure 12, with a maximum prediction error for images at different time steps is 3.34%. It is evident that the unsteady aerodynamic model has high generalisation capability, accurately predicting pitching oscillation flow fields beyond the training datasets.

4.3. Discussion on the generalization of the aerodynamic model based on transfer learning

In addition to the velocity field, the pressure field is also a key factor in the analysis of aerodynamic problems and has a significant impact on airfoil design. Although the velocity and pressure contours represent different physical quantities, they both originate from the same flow phenomena, and there is a certain correlation between them. The intelligent aerodynamic model constructed in this paper has learned useful features related to the flow, which can be partially applied to the pressure prediction task. Therefore, we further explored the generalisation of this intelligent aerodynamic model for pressure field prediction tasks based on transfer learning techniques.

The core idea of transfer learning is that many tasks share common features, especially within related fields.

Therefore, a model pre-trained on a large-scale dataset can be applied to other similar tasks through transfer learning. In this paper, this means applying the velocity field pre-trained model to the pressure field prediction task. In the intelligent aerodynamic model, shallow networks typically extract low-level, general flow features (such as edges, colours, textures, etc.), which are common for pressure field prediction tasks. Freezing is an important method in transfer learning. During training, by fixing the parameters of the shallow neural network and only updating the parameters of the layers close to the output, the general features from the pre-trained model can be fully utilised.

Given the complexity of unsteady aerodynamic models and their time-varying characteristics in the flow field, they involve more intricate dynamic behaviours. Therefore, the use of the unsteady aerodynamic model constructed in this paper for pressure field prediction will be a key focus of our future research, and further exploration and optimisation will be conducted. In this section, we primarily focus on discussing the steady-state intelligent aerodynamic model. Specifically, we freeze the shallow network parameters of the U-Net model and then fine-tune it on our small-scale pressure field dataset. The resulting pressure contour prediction is shown in the Figure 13, with the maximum error not exceeding 3.35%. The results demonstrate that the intelligent

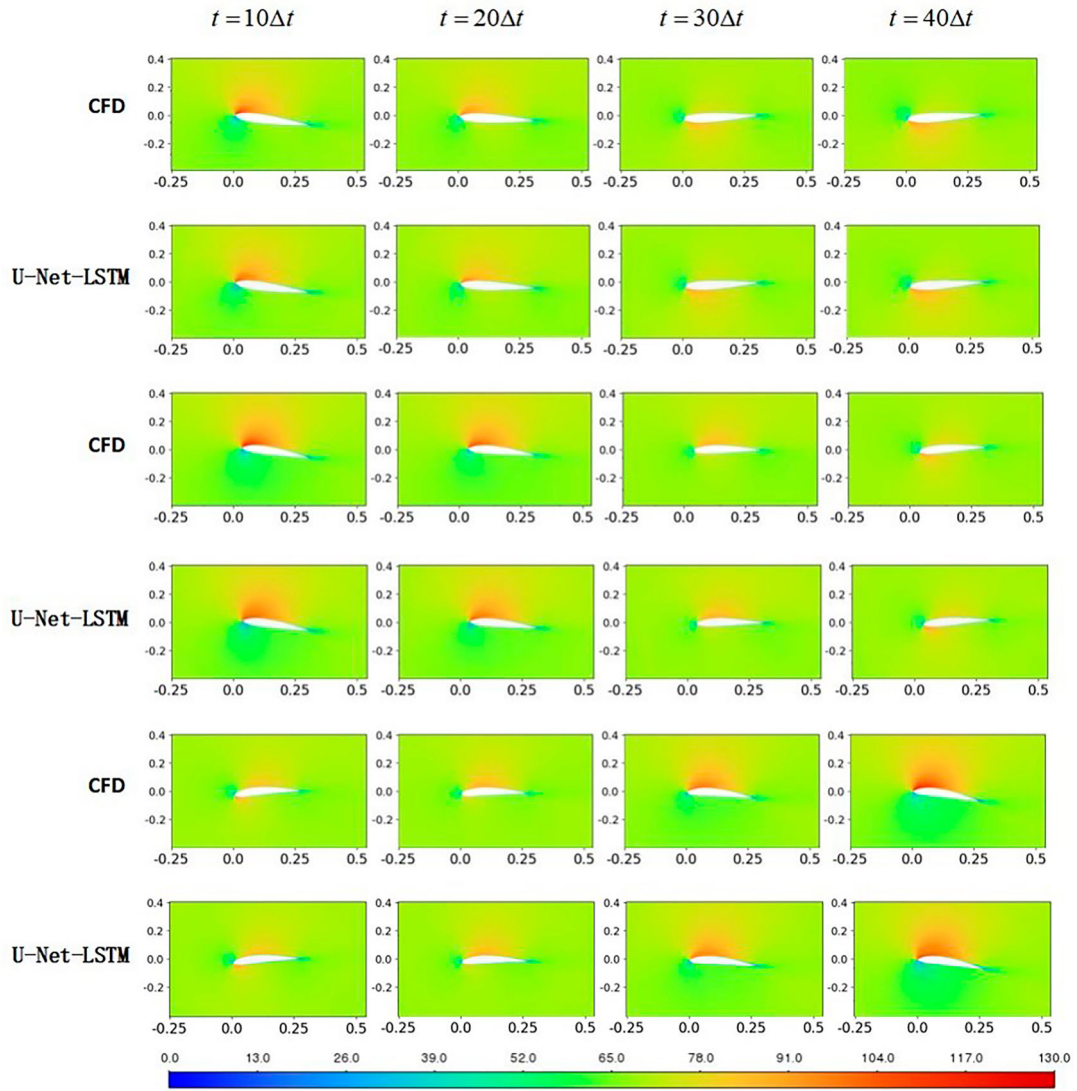


Figure 12. Ground truth and predicted results.

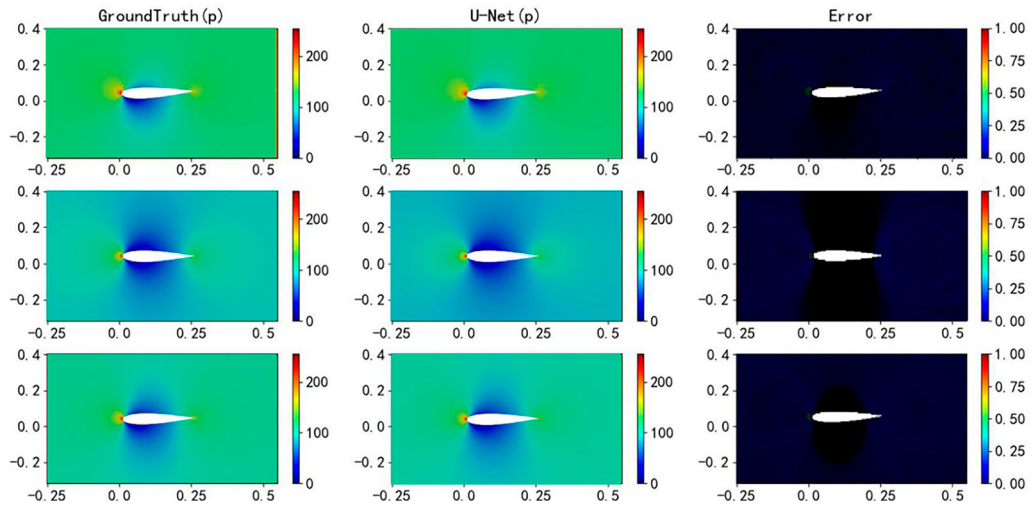


Figure 13. Ground truth and predicted results.

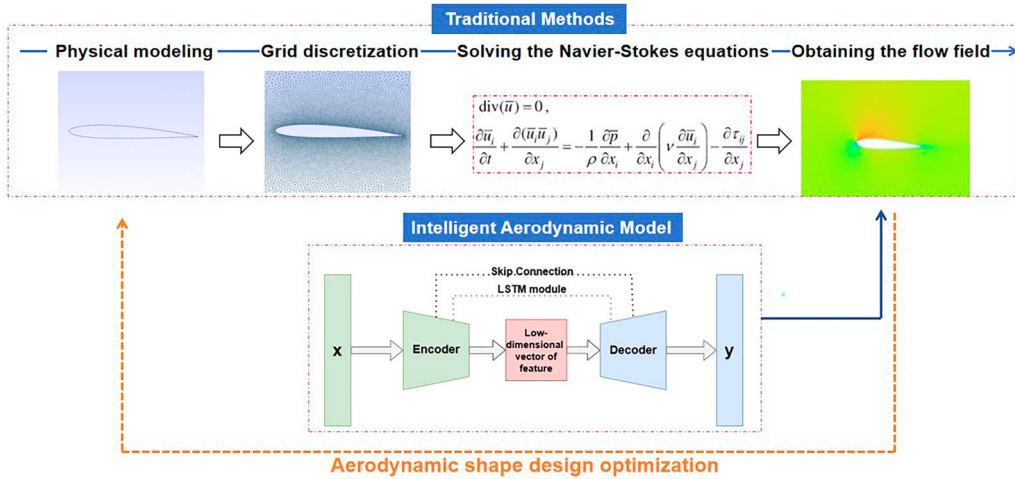


Figure 14. Iterative design of the airfoil.

aerodynamic model developed in this paper has strong generalisation and transferability, enabling accurate pressure field prediction by effectively utilising the general features extracted during the velocity field training process.

5. Conclusion

Efficient acquisition of the flow field around airfoils is crucial in aircraft design, especially in the early stages. To address the inefficiency in current methods for obtaining flow fields, an aerodynamic model was constructed using U-Net to rapidly predict the steady flow field around airfoils. Additionally, a Long Short-Term Memory (LSTM) module was incorporated to capture temporal information, enabling the rapid prediction of the unsteady flow field around airfoils. To address the problem of ‘dimension loss’ in the modelling datasets, effective data dimensionality enhancement was achieved using SDF images and ‘synthesized images’. The intelligent aerodynamic model proposed in this paper holds significant practical importance in the aerospace field, particularly during the early stages of aircraft design. By rapidly predicting the flow field around airfoils, this model provides engineers with real-time feedback during design iterations, helping them to promptly identify and address potential aerodynamic performance issues, as shown in Figure 14. Furthermore, in actual flight, the model can be utilised for real-time flow field prediction, assisting aircraft in adjusting flight control strategies in dynamic environments, thereby enhancing flight safety and performance. In addition to the aerospace sector, this intelligent aerodynamic model can also be extended to other areas, such as wind resistance testing in the automotive industry, by predicting the flow field around vehicles to optimise vehicle design and improve performance.

Verification analysis was conducted using the flow field around NACA airfoils and pitch oscillation flow fields, leading to the following conclusions:

1. By combining U-net and LSTM, and using SDF images along with ‘synthetic images’ to increase the dimensionality of airfoil geometry and computational conditions, an intelligent aerodynamic model for predicting the flow field around airfoils was constructed. This method can more intuitively represent the evolution and features of the flow field through images, demonstrating its physical significance to a certain extent.
2. The U-Net-LSTM intelligent model adapts well to the modelling and prediction of both steady and unsteady flow fields. The U-Net model extracts spatial features from flow field images, while the LSTM module captures temporal features, resulting in more accurate prediction of the flow field. The maximum prediction error for the steady flow field does not exceed 1.98%, and for the unsteady flow field, the prediction error does not exceed 2.56%.
3. Since images contain multi-level and multi-dimensional features (including changes in colour and brightness, geometric shapes, and boundary information), combining them with the aerodynamic model proposed in this paper allows for the learning of more intrinsic rules. Therefore, the aerodynamic model presented in this paper demonstrates good generalisation in predicting both steady and unsteady flow field.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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